STUDENT PERFORMANCE ANALYSIS AND PREDICTION SYSTEM

MINOR PROJECT REPORT

Submitted as part of

AI/ML WITH PYTHON TRAINING PROGRAM

Organised by InternsElite

SUBMITTED BY

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PROJECT DOMAIN

Artificial Intelligence and Machine Learning

PROGRAMMING LANGUAGE

Python

BATCH

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ACADEMIC YEAR

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1. INTRODUCTION

Student performance is influenced by several factors, including study habits, previous academic results, attendance and personal circumstances. Understanding these factors can help identify patterns in academic achievement.

This project, titled Student Performance Analysis and Prediction System, uses Python and machine learning techniques to study student performance. The analysis is based on a dataset obtained from the UCI Machine Learning Repository, containing information about students and their academic records.

The project mainly focuses on the final mathematics grade (G3). Data analysis and visualisation are used to understand patterns in the dataset, while a Linear Regression model is developed to predict students' final grades using selected academic and personal features.

The purpose is to explore how data analysis and machine learning can help in understanding academic performance. The predictions are intended for educational analysis and should not be treated as a final judgement of a student's abilities.

2. OBJECTIVES OF THE PROJECT

The main objectives of this project are:

1. To study and understand the factors that may influence students' academic performance.

2. To explore the student performance dataset using Python and statistical methods.

3. To clean and prepare the dataset for further analysis.

4. To create visualisations such as histograms, bar charts, scatter plots and correlation heatmaps to identify patterns.

5. To develop a Linear Regression model for predicting students' final mathematics grades.

6. To evaluate the model using suitable performance metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R^2 score.

7. To understand the usefulness and limitations of machine learning in analysing educational data.

3. DATASET DESCRIPTION

The dataset used in this project is the Student Performance Dataset, obtained from the UCI Machine Learning Repository.

It contains information about students' demographic details, family background, study habits and academic performance. The dataset used for this project is the mathematics student record file ("student-mat.csv").

Dataset details:

- Source: UCI Machine Learning Repository
- File name: student-mat.csv
- Number of records: 395 students
- Number of attributes: 33
- Target variable: G3 (final mathematics grade)

Selected Features

For this project, six features were selected for model development:

1. "studytime" – Weekly study-time category.
2. "failures" – Number of past class failures.
3. "absences" – Number of school absences.
4. "G1" – First-period grade.
5. "G2" – Second-period grade.
6. "age" – Age of the student.

The target variable, "G3", represents the student's final mathematics grade. These features were selected to examine how academic history, study habits and other student-related factors relate to final performance.

4. TOOLS AND TECHNOLOGIES USED

The project was developed using Python and its data analysis and machine learning libraries. The following tools and technologies were used:

1. Python

Python was used as the main programming language for data processing, visualisation and model development.

2. Google Colab

Google Colab provided the online environment for writing and executing Python code without installing software locally.

3. Pandas

Pandas was used to load, organise, clean and manipulate the student dataset.

4. NumPy

NumPy supported numerical operations and data handling.

5. Matplotlib and Seaborn

These libraries were used to create graphs and visualisations for exploring the dataset.

6. Scikit-learn

Scikit-learn was used to split the dataset, develop the Linear Regression model, evaluate its performance and perform cross-validation and model tuning.

7. Joblib

Joblib was used to save and reload the trained model and its feature information.

5. DATA CLEANING AND PREPROCESSING

Before performing the analysis, the dataset was prepared to make it suitable for model development.

The following steps were carried out:

1. Loading the Dataset

The "student-mat.csv" file was loaded into Python using the Pandas library.

2. Understanding the Data

The dataset was examined to understand its structure, number of rows and columns, and the types of information it contained.

3. Checking Missing Values

The dataset was checked for missing values. The "dropna()" method was applied to remove rows containing missing values, if present.

4. Removing Duplicate Records

Duplicate records were checked and the "drop_duplicates()" method was used to remove repeated entries.

5. Feature Selection

Six input features were selected: studytime, failures, absences, G1, G2 and age. The final grade (G3) was selected as the target variable.

6. Preparing Data for Modelling

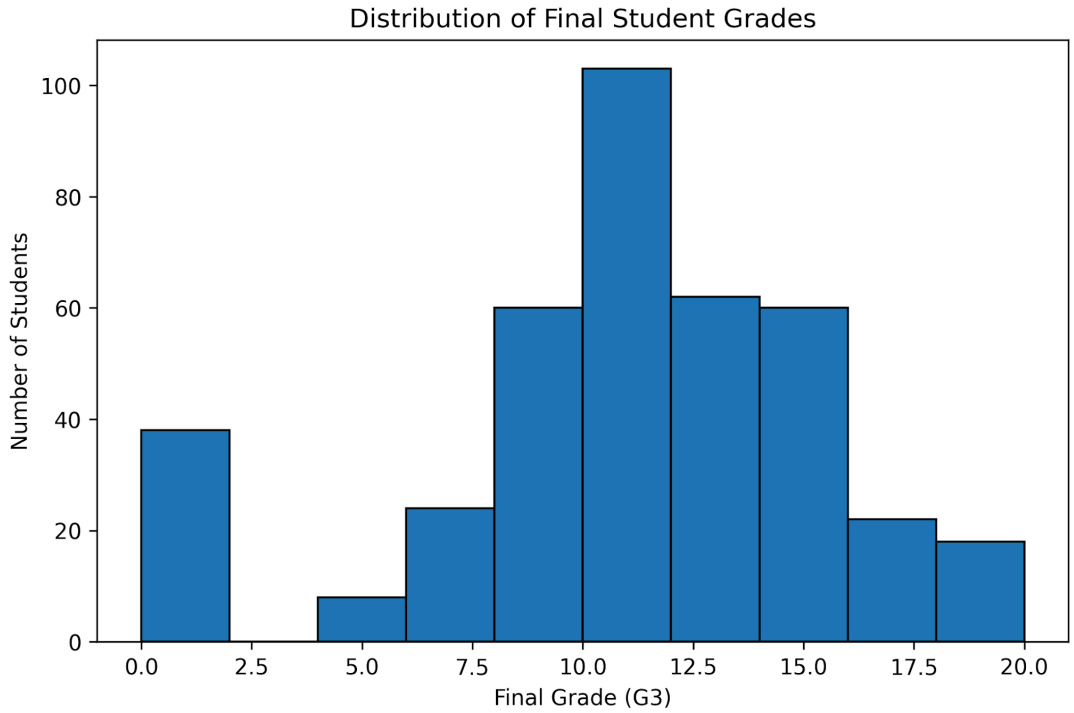
The selected input features were stored separately from the target variable. The data was then divided into training and testing sets using an 80:20 ratio. This allowed the model to be trained on one portion of the data and evaluated on unseen data.

6. EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis was performed to understand the distribution of student grades and identify relationships between different variables. Five visualisations were created using Matplotlib and Seaborn.

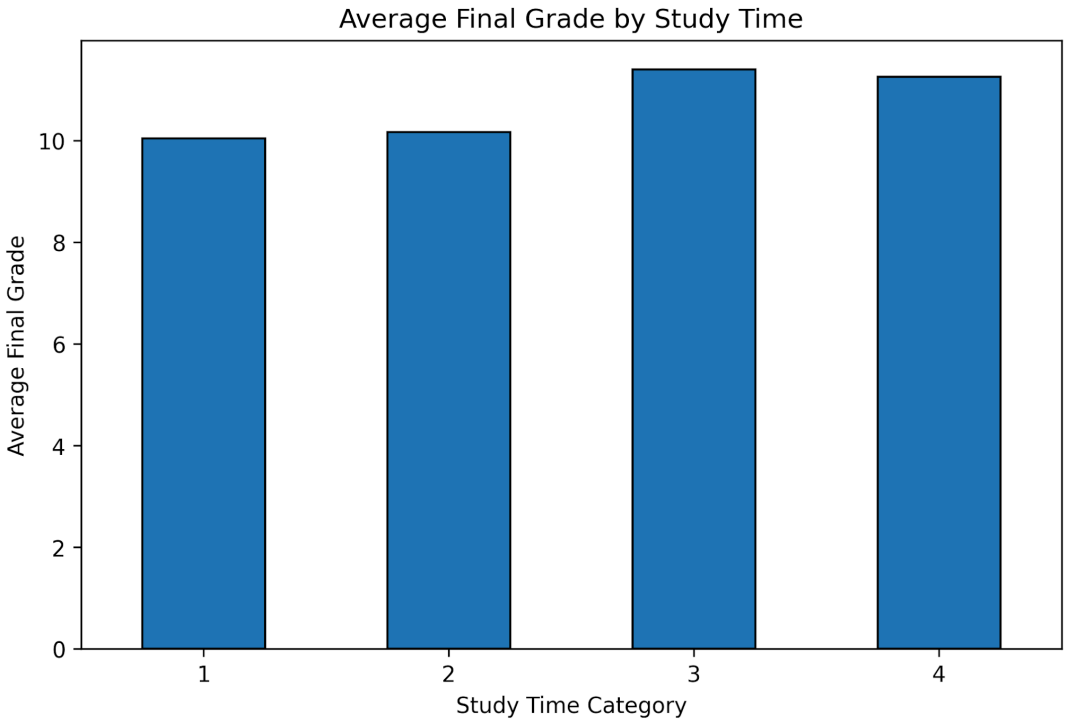
1. Histogram of Final Grades (G3)

The histogram shows the distribution of students' final mathematics grades. It helps understand how frequently different grade values occur and gives an overview of students' performance.



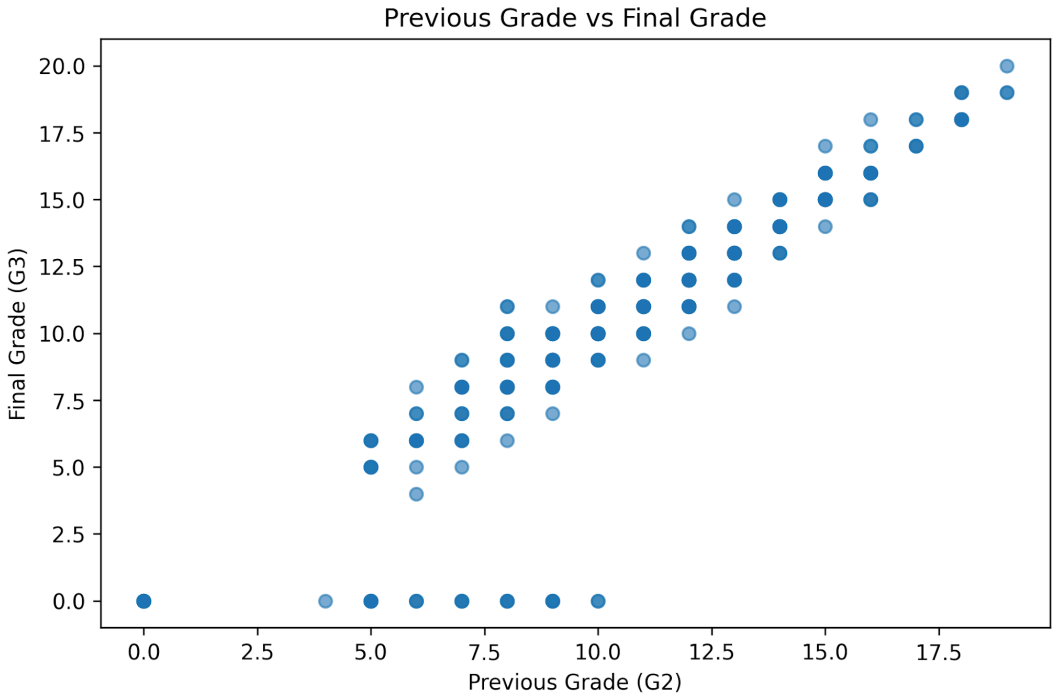
2. Bar Chart – Study Time and Final Grade

This chart compares the average final grade (G3) across different study-time categories. It helps examine whether students with different study-time patterns show differences in academic performance.



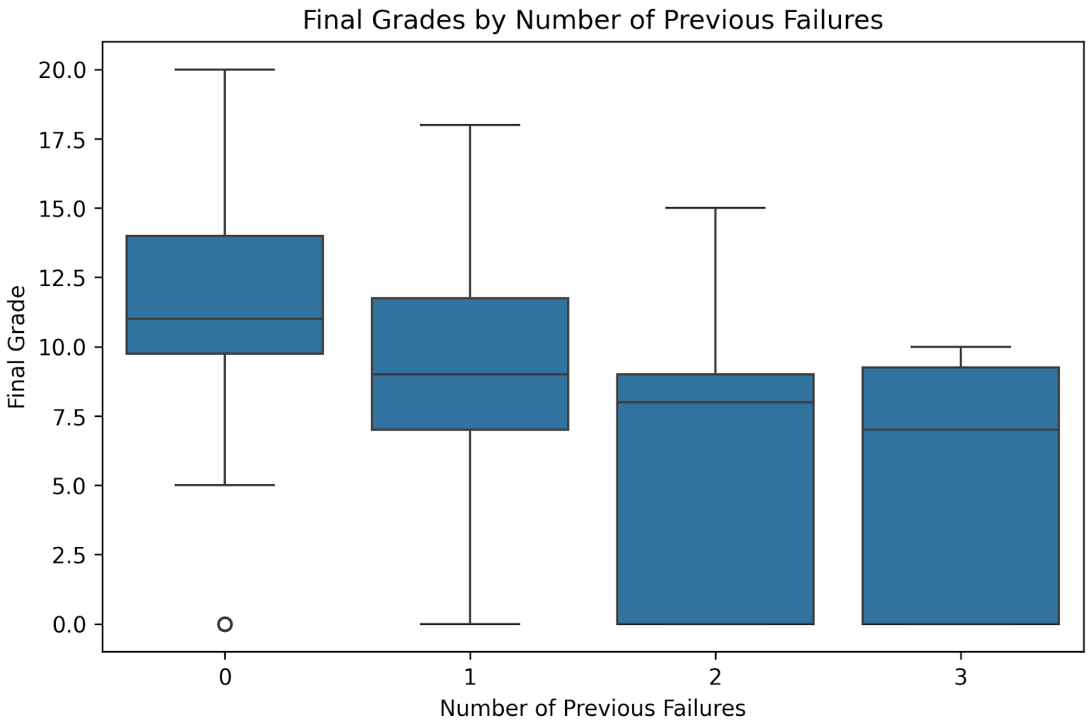
3. Scatter Plot – G2 and G3

The scatter plot represents the relationship between second-period grades (G2) and final grades (G3). It helps observe whether students who performed well in the second period also tended to achieve higher final grades.



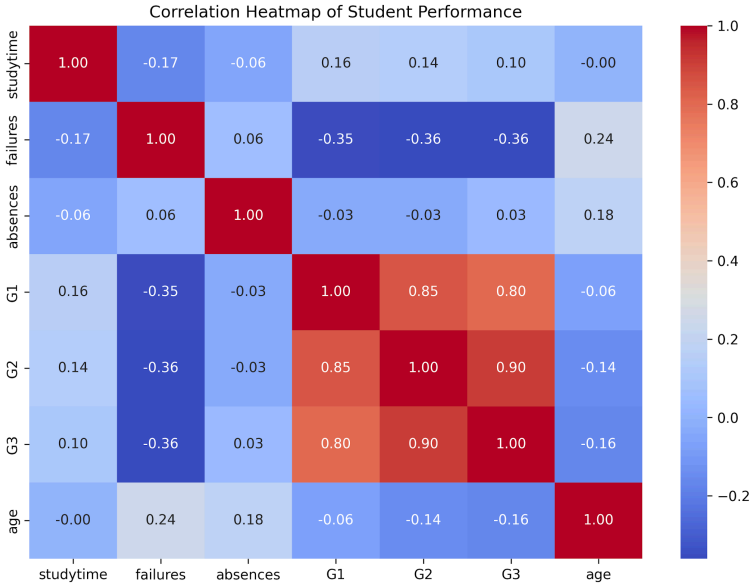
4. Box Plot – Failures and Final Grade

The box plot compares final mathematics grades across different numbers of past class failures. It helps examine variations in performance among these groups.



5. Correlation Heatmap

The heatmap presents the correlation between selected numerical variables. It helps identify the strength and direction of relationships between the features and the final grade.



Overall Observation

These visualisations provide a clearer understanding of the dataset and help identify patterns that can be explored further through machine learning.

7. MODEL DEVELOPMENT

After completing data cleaning and exploratory analysis, a Linear Regression model was developed to predict students' final mathematics grades.

1. Selection of Input and Target Variables

Six features were selected as input variables:

- Study time
- Past class failures
- Absences
- First-period grade (G1)
- Second-period grade (G2)
- Age

The final mathematics grade (G3) was used as the target variable.

2. Splitting the Dataset

The dataset was divided into two parts:

- 80% for training the model.
- 20% for testing the model.

The "train_test_split()" function from Scikit-learn was used, with a random state of 42 to maintain reproducibility.

3. Training the Model

The Linear Regression algorithm was applied to the training data. The model learned the relationship between the selected input features and the final mathematics grade.

4. Making Predictions

After training, the model was used to predict the final grades for the test dataset. These predicted values were then compared with the actual grades to evaluate model performance.

The model was saved using Joblib so that it could be loaded and reused without training it again.

8. MODEL EVALUATION

After training the Linear Regression model, its performance was evaluated using three commonly used evaluation metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R^2 Score.

1. Mean Absolute Error (MAE)

MAE measures the average absolute difference between the actual grades and the grades predicted by the model.

Obtained Value: 1.33

This means that the model's predictions differed from the actual grades by approximately 1.33 grade points on average.

2. Root Mean Squared Error (RMSE)

RMSE measures the prediction error while giving greater weight to larger errors.

Obtained Value: 2.11

3. R^2 Score (Coefficient of Determination)

The R^2 score indicates how well the model explains variations in the target variable.

Obtained Value: 0.782

This indicates that the model explains approximately 78.2% of the variation in final mathematics grades within the test dataset.

Evaluation Summary

Evaluation Metric	**Obtained Value**

MAE	1.33
RMSE	2.11
R^2 Score	0.782

Overall, the evaluation results indicate that the Linear Regression model was able to identify useful patterns in the data and produce reasonably accurate predictions. However, its performance may vary when applied to other student datasets.

9. CROSS-VALIDATION AND MODEL TUNING

To further examine the model's performance, cross-validation and basic model tuning were performed using Scikit-learn.

1. Cross-Validation

Five-fold cross-validation was applied to evaluate how consistently the model performed across different portions of the dataset.

The dataset was divided into five parts, and the model was trained and evaluated across different combinations of these parts.

- Number of folds: 5
- Average R^2 Score: 0.819
- Standard Deviation: 0.031

The results indicate relatively consistent performance across the five folds.

2. Model Tuning Using Ridge Regression

Ridge Regression was also tested as an alternative to Linear Regression. It uses regularisation to reduce the influence of large model coefficients.

A GridSearchCV procedure was used to test different alpha values:

0.01, 0.1, 1, 10 and 100.

The best alpha value obtained was 0.01, with a cross-validation R^2 score of 0.834.

3. Comparison of Results

Although Ridge Regression achieved a higher cross-validation score, its test-set R^2 score was 0.782, the same as the original Linear Regression model.

Therefore, Linear Regression was retained as the main model for this project.

This comparison shows that a higher cross-validation score does not necessarily result in better performance on the test dataset.

10. RESULTS AND DISCUSSION

The project successfully analysed the Student Performance Dataset and developed a machine learning model to predict students' final mathematics grades.

The major findings of the project are summarised below:

1. Dataset Analysis

Exploratory Data Analysis helped examine the distribution of final grades and the relationships between study time, previous grades, past failures and other selected variables.

2. Model Performance

The Linear Regression model achieved the following results on the test dataset:

- MAE: 1.33
- RMSE: 2.11
- R^2 Score: 0.782

These results suggest that the model was able to capture a considerable amount of variation in students' final grades.

3. Cross-Validation Results

The five-fold cross-validation produced an average R^2 score of 0.819, with a standard deviation of 0.031. This suggests reasonably consistent performance across the different folds.

4. Model Comparison

Ridge Regression was tested through basic hyperparameter tuning. Although it achieved a higher cross-validation score, it did not improve the test-set R^2 score over Linear Regression..

Overall Discussion

The analysis demonstrates how Python and machine learning can be used to explore academic data and estimate student performance. However, the predictions depend on the available dataset and selected features. They should be treated as analytical estimates rather than exact outcomes.

11. LIMITATIONS OF THE PROJECT

Although the project provides useful insights into student performance, it has certain limitations:

1. Limited Dataset: The analysis is based on a dataset containing 395 student records, which may not represent students from all educational backgrounds.

2. Limited Features: Only six input features were selected for prediction. Other factors, such as teaching methods, learning environment and personal circumstances, were not included.

3. Prediction Accuracy: The model provides estimates rather than exact predictions. Actual academic performance may differ from predicted grades.

4. Dataset Context: The dataset comes from a particular educational setting, so the findings may not apply equally to all schools or student groups.

5. Model Limitations: Linear Regression assumes a linear relationship between input features and the target variable. It may not capture every complex pattern in student performance.

Despite these limitations, the project provides practical experience in data analysis, visualisation and machine learning.

12. CONCLUSION

The project, Student Performance Analysis and Prediction System, provided an opportunity to understand how data analysis and machine learning can be applied to educational data.

Using Python, the Student Performance Dataset was cleaned, explored and visualised to identify patterns related to students' academic performance. Five visualisations were created to examine grade distribution and relationships between selected variables.

A Linear Regression model was developed to predict final mathematics grades. It achieved an R^2 score of 0.782, with an MAE of 1.33 and an RMSE of 2.11 on the test dataset.

The project also introduced cross-validation and Ridge Regression tuning, helping compare model performance and understand the importance of evaluation.

Overall, this project improved my understanding of Python programming, data handling, visualisation and basic machine learning. It also showed that machine learning can support educational data analysis, provided its predictions are interpreted carefully and its limitations are considered.

13. REFERENCES

The following sources and documentation were referred to during the development of this project:

1. UCI Machine Learning Repository. Student Performance Dataset.
   https://archive.ics.uci.edu/dataset/320/student+performance

2. Python Documentation. Python Programming Language.
   https://docs.python.org/3/

3. Pandas Documentation. Data Analysis and Manipulation Library.
   https://pandas.pydata.org/docs/

4. NumPy Documentation. Numerical Computing Library.
   https://numpy.org/doc/

5. Matplotlib Documentation. Data Visualisation Library.
   https://matplotlib.org/stable/

6. Seaborn Documentation. Statistical Data Visualisation.
   https://seaborn.pydata.org/

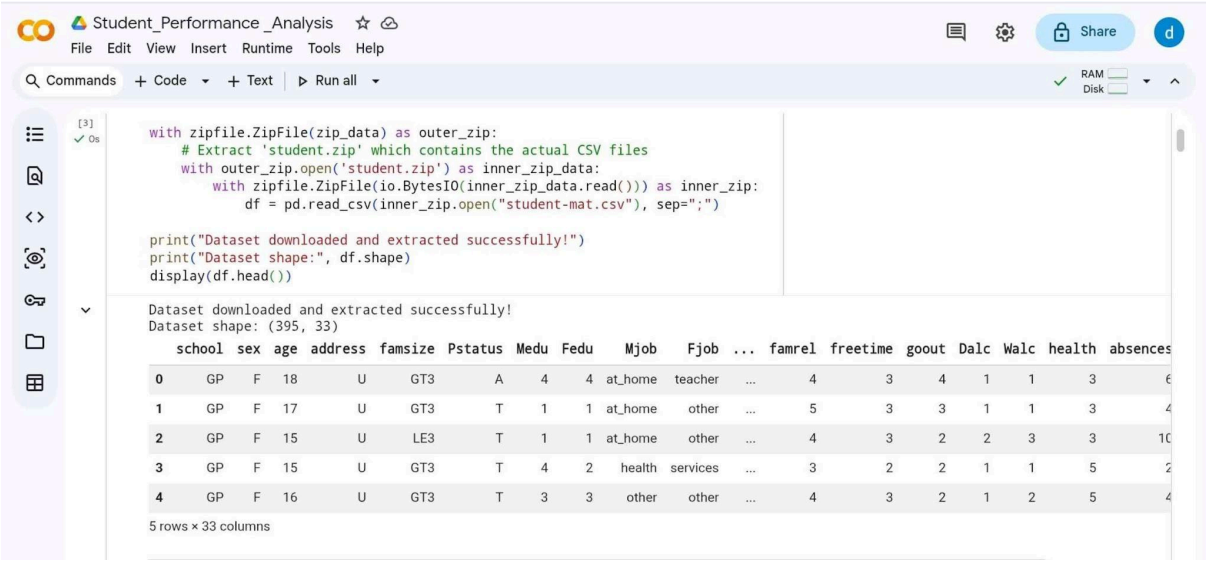
7. Scikit-learn Documentation. Machine Learning Tools and Algorithms.
   https://scikit-learn.org/stable/

8. Joblib Documentation. Model Persistence and Utility Functions.
   https://joblib.readthedocs.io/

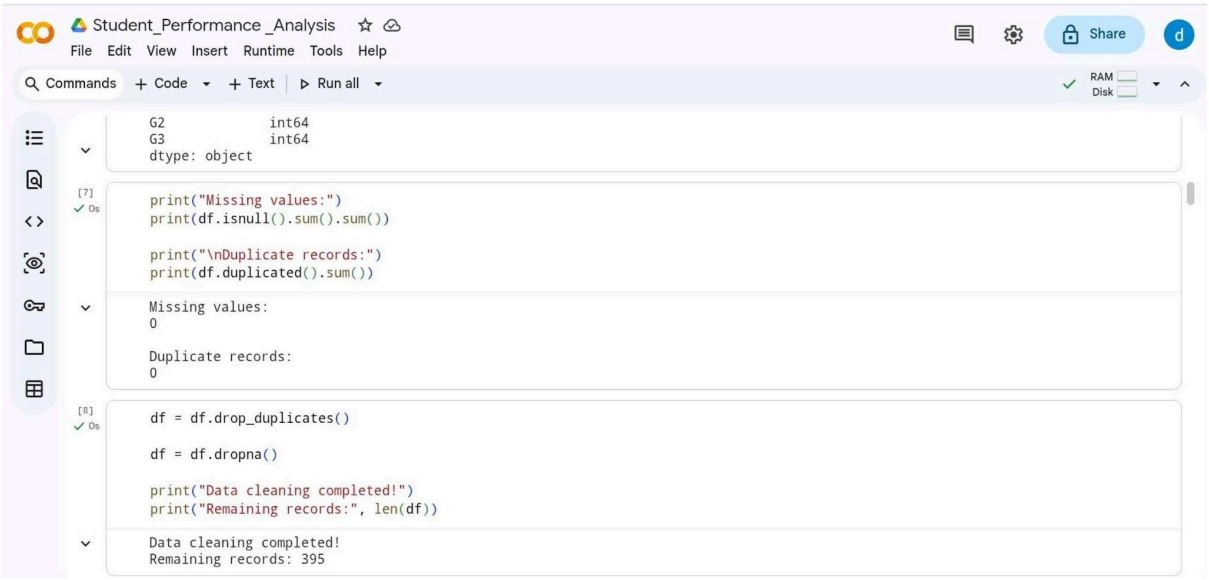
APPENDIX A: PROJECT IMPLEMENTATION SCREENSHOTS

The following screenshots provide practical evidence of the implementation of the Student Performance Analysis and Prediction System using Python in Google Colab.

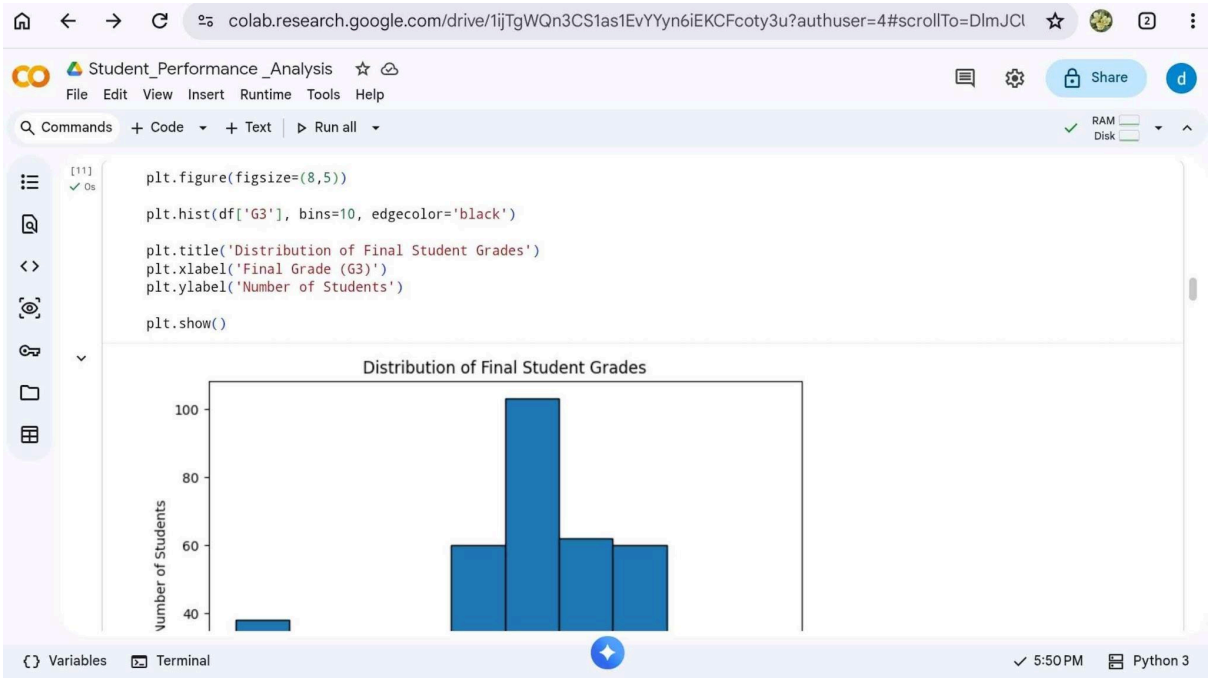
- Figure A1: Loading and inspecting the student dataset.



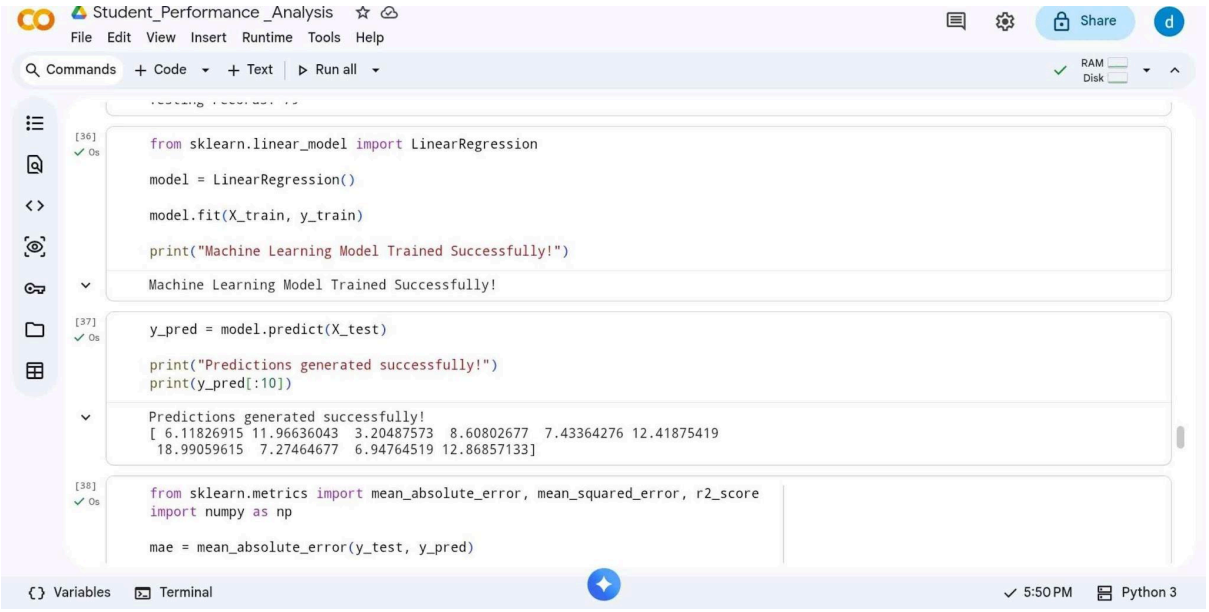
- Figure A2: Checking missing values and duplicate records.



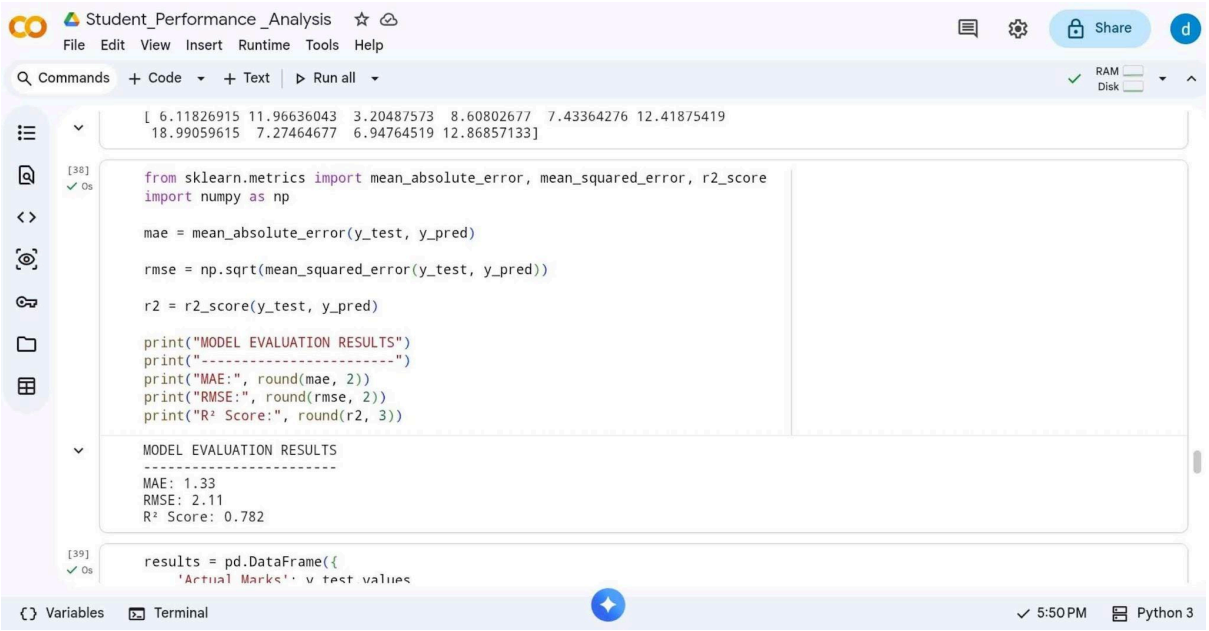
- Figure A3: Generating visualisations using Python.



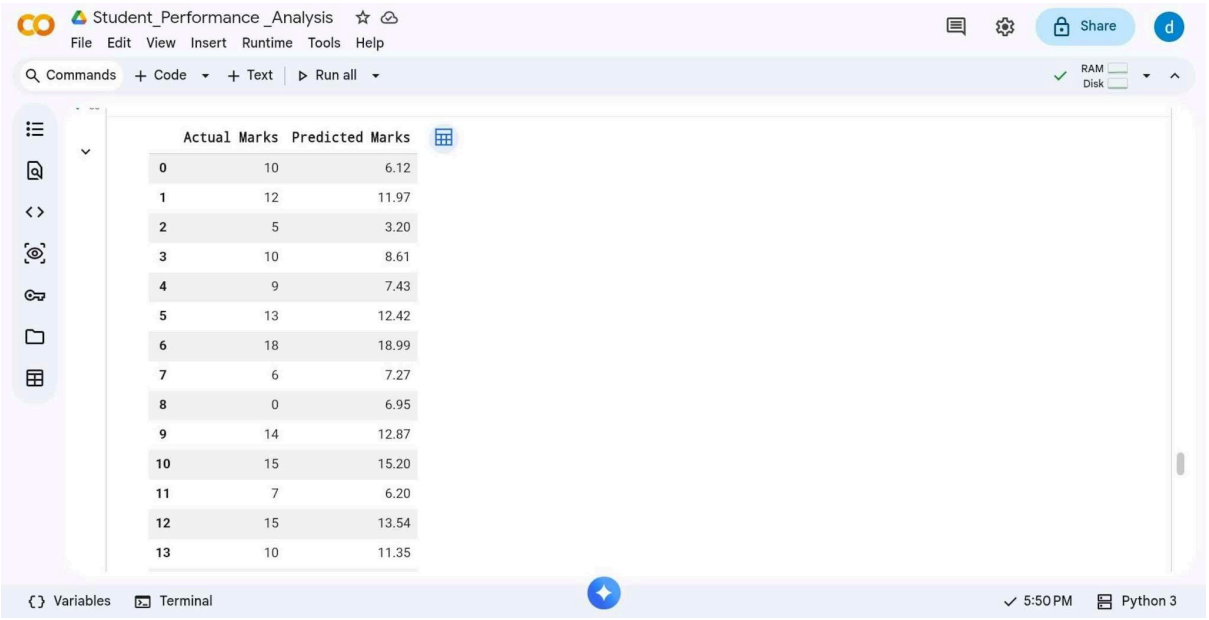
- Figure A4: Training the Linear Regression model.



- Figure A5: Evaluating model performance.



- Figure A6: Comparing actual and predicted grades.



- Figure A7: Saving and reloading the trained model.

